
CHAPTER 5

COMMON PARAMETRIC DISCRETE RANDOM VARIABLES

DISCRETE UNIFORM DISTRIBUTION

BERNOULLI DISTRIBUTION

BINOMIAL DISTRIBUTION

POISSON DISTRIBUTION

MOTIVATION

- Many real-life experiments follow similar patterns:
 - Experiments with only two possible outcomes ('success', 'failure')
 - Experiments that involve a sequence of binary outcomes
 - Experiments that involve sampling without replacement
 - Experiments that involve counting occurrences over a given space or time
- These random variables arising from these common experiments merit more formal pmfs and properties
- These distributions are governed by parameters, which govern the pmf, mean, variance, etc.



DISCRETE UNIFORM DISTRIBUTION



MOTIVATION

- Consider the following:
 - Rolling a fair dice. Let X = the number of dots on the top of the dice.
 - $X = 1, 2, 3, 4, 5, 6$
 - Flipping a fair coin once. Let X = the number of heads.
 - $X = 0, 1$
- Each outcome has an equal probability of occurring.

DISCRETE UNIFORM DISTRIBUTION

- A random variable X has a **discrete uniform distribution** if each of the n values in its range, say $x_1, x_2, \dots, x_n$ has equal probability.

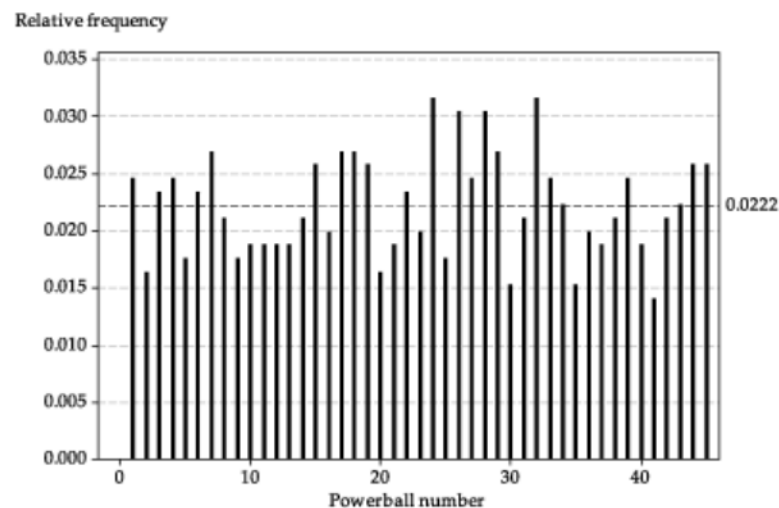
- We say $X \sim \text{DISUNIF}(n)$ where the pmf is given by:

$$P(X = x) = f(x; n) = \begin{cases} \frac{1}{n}, & x = x_1, x_2, \dots, x_n \\ 0, & \text{otherwise} \end{cases}$$

EXAMPLE

- In commercial lotteries, such as Powerball, it is a regulatory requirement that each outcome is equally likely. There are 45 possible Powerball numbers (1, 2, ..., 45). Let X be the Powerball drawn on a particular occasion. This is a discrete uniform distribution and the probability for each of the 45 possible values is:

$$P(X = x) = f(x) = \frac{1}{45} \approx 0.0222.$$



The graph shows the relative frequencies observed in the 853 draws from May 1996 to September 2012.

Relative frequencies of Powerball numbers over 853 draws.

DISCRETE UNIFORM DISTRIBUTION – MEAN & VARIANCE

- Suppose X is a discrete uniform random variable on the consecutive integers $a, a + 1, a + 2, \dots, b$ for $a \leq b$.

- The mean of X is

$$\mu = E[X] = \frac{a + b}{2}$$

- The variance of X is

$$\sigma^2 = \text{Var}(X) = \frac{(b - a + 1)^2 - 1}{12}$$

- Note: if you compute the mean and variance by their definitions (i.e. using the possible x -values from a to b , $f(x) = \frac{1}{n}$, $E[X] = \sum x \cdot f(x)$), you will derive the above formulas.

EXAMPLES

- Roll a fair coin twice and observe the top face. Let X be the number of HEADS occurred. Is X a discrete uniform random variable?
- Roll a fair die once and observe the top face. Let X be the number of dots observed. Is X a discrete uniform random variable?

EXAMPLE I

- Roll a fair die once and observe the top face. Let X be the number of dots observed.
 - State the distribution name and parameter(s).
 - Specify the pmf.
- Find $E[X] =$
- Find $\text{Var}(X) =$
- Find $P(X < 3) =$
- Find $P(X \leq 3) =$



BERNOULLI DISTRIBUTION



MOTIVATION

- Consider the following experiments:
 - Flip a coin. Observe whether it lands heads or tails.
 - Shoot a basketball. Observe whether you make or miss.
 - Hire a new CEO. Observe whether the individual is male or female.
 - Roll a roulette wheel. Observe whether it lands red or not.

MOTIVATION

- All of these experiments have binary outcomes: 1/0, or ‘Success’/ ‘Failure’
- We let $X = 1$ if success, $X = 0$ if failure
- Then:

$$P(X = x) = \begin{cases} p, & x = 1 \\ 1 - p, & x = 0 \\ 0, & \text{otherwise} \end{cases}$$

BERNOULLI DISTRIBUTION

- This is known as the Bernoulli distribution
- We say that $X \sim \text{BERN}(p)$ if the pmf of X is:

$$P(X = x) = f(x; p) = \begin{cases} p^x(1-p)^{1-x}, & x = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

- Verify that this matches the pmf on the previous slide

BERNOULLI DISTRIBUTION – EXPECTED VALUE / VARIANCE

- Suppose $X \sim \text{BERN}(p)$

- The mean of X is

$$\mu = E[X] =$$

- The variance of X is

$$\sigma^2 = \text{Var}(X) =$$

EXAMPLE 2

- Experiment: LeBron James takes one free throw, 1 = make; 0 = miss.
- His career free throw shooting percentage is 73.7%.
 - State the distribution name and parameter(s).
 - Specify the pmf.
- Find $E[X] =$
- Find $\text{Var}(X) =$

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BINOMIAL DISTRIBUTION



MOTIVATION

- Many experiments are comprised of sequences of n independent Bernoulli experiments
- If the random variable X is the number of “successes” in n independent trials, then X is said to follow a **binomial distribution**.
- Examples:
 - Number of heads out of 10 coin flips
 - Number of free throws made out of 20 shots taken
 - Number of O+ people in a sample of 50

BINOMIAL DISTRIBUTION

- We say $X \sim \text{BIN}(n, p)$ if $X = \sum_{i=1}^n Y_i$, where Y_i are independent $\text{BERN}(p)$ trials
- Note from definition, X = number of successes in n independent Bernoulli trials
- The pmf is given by

$$p(x) = P(X = x) = f(x; n, p) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x}, & x = 0, 1, 2, \dots, n \\ 0 & \text{otherwise} \end{cases}$$

EXAMPLE 3

- Let X = number of 1's rolled.
 - Suppose you roll two dice. State the distribution name and parameter(s).
 - Suppose you roll two dice. Specify the pmf.
- Using the binomial distribution, find $P(X = \text{one } 1 \text{ is rolled})$ when rolling two dice.

EXAMPLE 3

- Let X = number of 1's rolled.
 - Now suppose you roll three dice. How does your distribution change?
- Using the binomial distribution, find $P(X = \text{two 1's are rolled})$ when rolling three dice.
- Using the binomial distribution, find $P(X = \text{three 1's are rolled})$ when rolling three dice.

BINOMIAL DISTRIBUTION

- The binomial distribution can be used when the following assumptions/conditions have been satisfied.
 - There are a fixed number of trials, N .
 - There are only two possible outcomes: a “success” and a “failure.”
 - $P(\text{success}) = p$ for each trial.
 - The trials are independent of one another.

EXAMPLE 4

- It is known that screws produced by a certain company will be defective with probability 0.01, independent of each other. The company sells screws in packages of 10 and offers a money-back guarantee that at least 2 of the 10 screws are defective. Let X = the number of defective screws in a package.
 - Is X a binomial random variable? Justify.
 - Find the probability that a package of screws will have to be replaced.

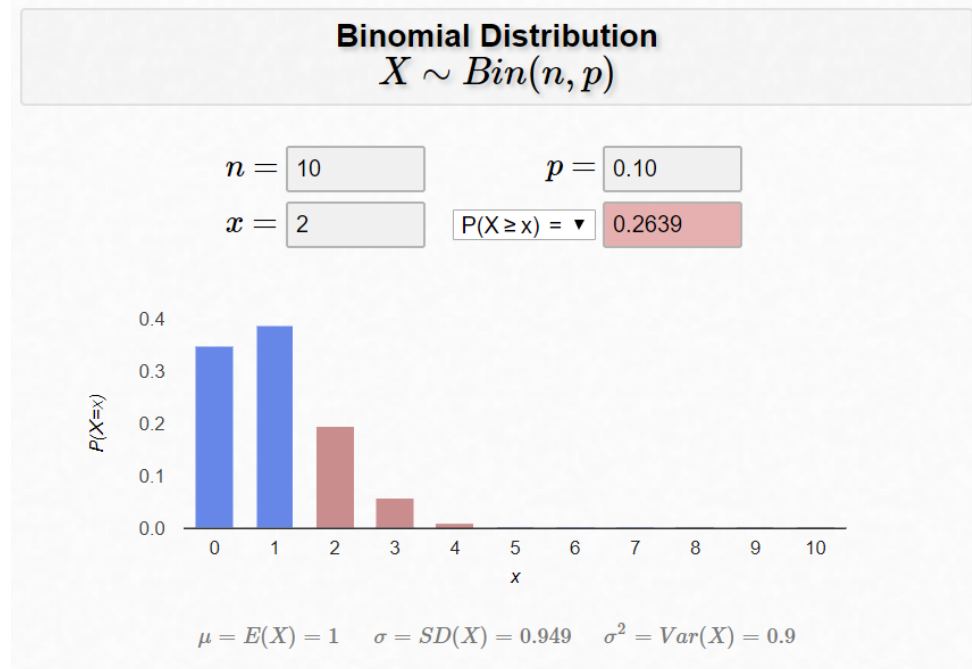
TECHNOLOGY

In calculator:

- $1 - \text{binompdf}(10, 0.1, 0) - \text{binompdf}(10, 0.1, 1)$

or

- $1 - \text{binomcdf}(10, 0.1, 1)$



EXAMPLE 5

- On a multiple choice exam with 3 possible answers for each of 5 questions, what is the probability that a student will get 4 or more correct answers just by guessing?
 - Hint: What are your parameters?

BINOMIAL DISTRIBUTION – EXPECTED VALUE / VARIANCE

- Bernoulli $\rightarrow$ one trial
 - $E[X] = p$ and $\text{Var}(X) = p(1 - p)$
- Binomial $\rightarrow n$ trials
- Recall that $X_1, X_2, \dots, X_n$ be independent Bernoulli trials and $Y = \sum_{i=1}^n X_i$. Find $E[Y]$ and $V(Y)$.

BINOMIAL DISTRIBUTION – EXPECTED VALUE / VARIANCE

- Suppose $X \sim \text{BIN}(n, p)$

- The mean of X is

$$\mu = E[X] = np$$

- The variance of X is

$$\sigma^2 = \text{Var}(X) = np(1 - p)$$

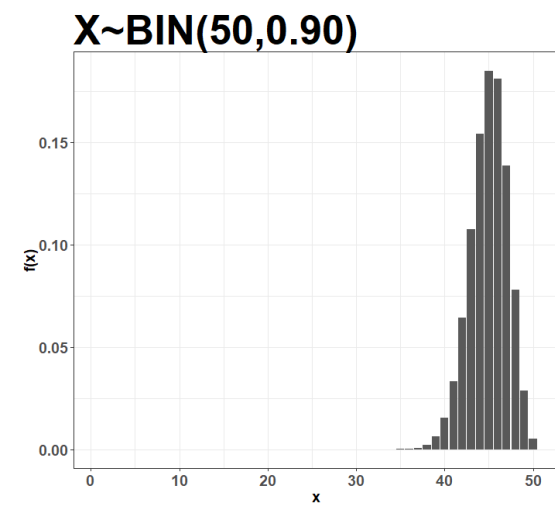
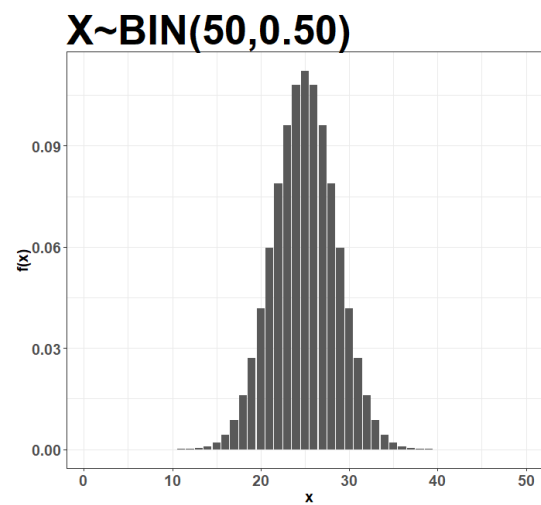
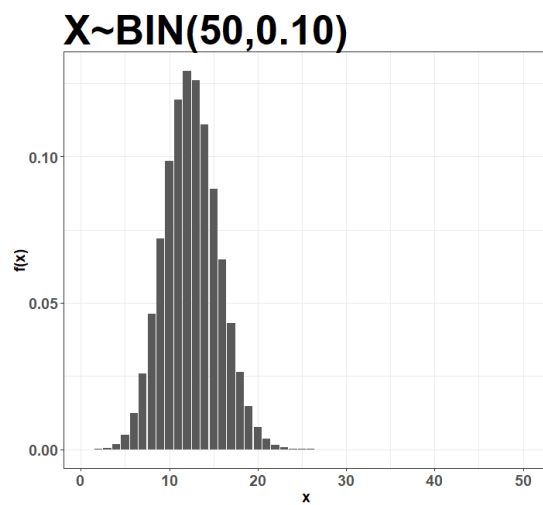
EXAMPLE 6

- Find the mean and variance for the RV X given in Example 4 (defective screws). Recall: $X \sim \text{BIN}(10, 0.01)$. What does $E[X]$ tell us in context of the situation?

- Find the mean and variance for the RV X given in Example 5 (multiple choice quiz). Hint: What are your parameters? What does $E[X]$ tell us in context of the situation?

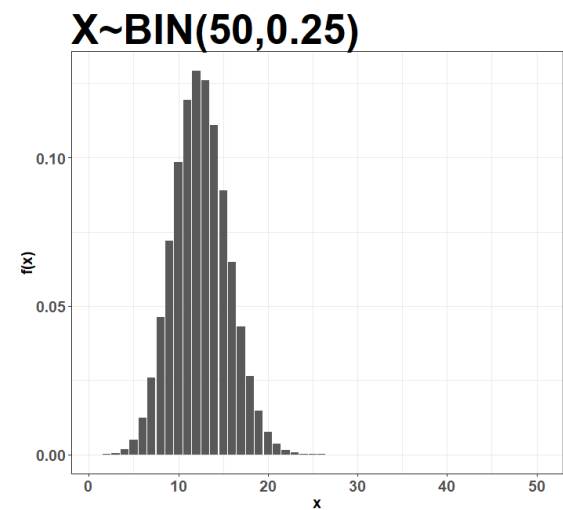
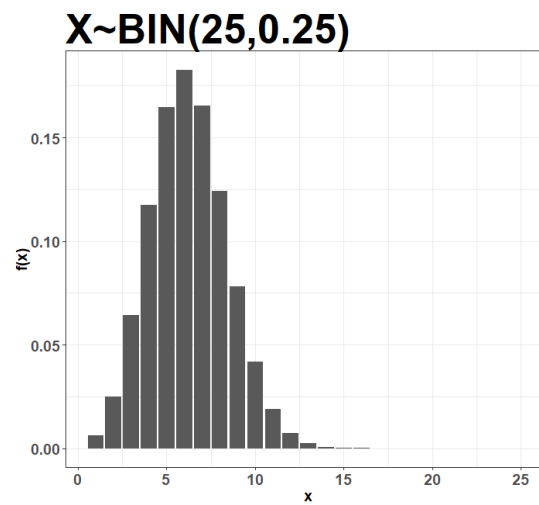
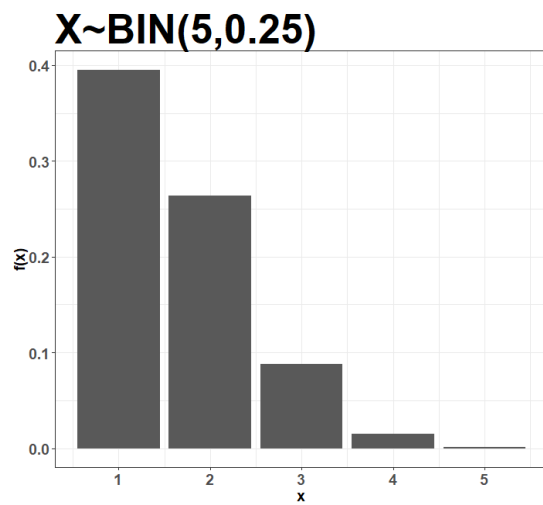
BINOMIAL DISTRIBUTION

- What happens as p increases?



BINOMIAL DISTRIBUTION

- What happens if n increases?



IN-CLASS

- If sampling 100 citizens from a city with 60% mayor approval rating, what is the probability the sample will have between 55 and 65 people who approve of the mayor? (not inclusive)



POISSON DISTRIBUTION



MOTIVATION

- Consider experiments that measure counts per unit space or time
- Examples
 - Number of cars at a drive-thru window in one hour
 - Number of accidents along a 0.5-mile stretch of highway
 - Number of tadpoles per cubic cm of pondwater
- These random variables are often modeled with the **Poisson distribution**

THE POISSON DISTRIBUTION

- Let X be the number of events in a Poisson process over a fixed interval of time or space.
- We say $X \sim \text{POIS}(\lambda)$ with pmf given by:

$$p(x) = P(X = x) = f(x; \lambda) = \begin{cases} \frac{\lambda^x e^{-\lambda}}{x!} & x = 0, 1, \dots \\ 0 & \text{otherwise} \end{cases}$$

- The parameter λ is the rate of events per unit time or space

EXAMPLE 7

- Consider a telemarketer who, on average, handles 5 calls every three minutes. Suppose X = the number of phone calls received in a minute and we're interested in finding the probability there will be no calls in the next minute.
 - State the distribution name and parameter's.
 - Specify the pmf.
 - Let's find the probability there will be no calls in the next minute.
 - Find the probability there will be at least 2 calls in the next minute.

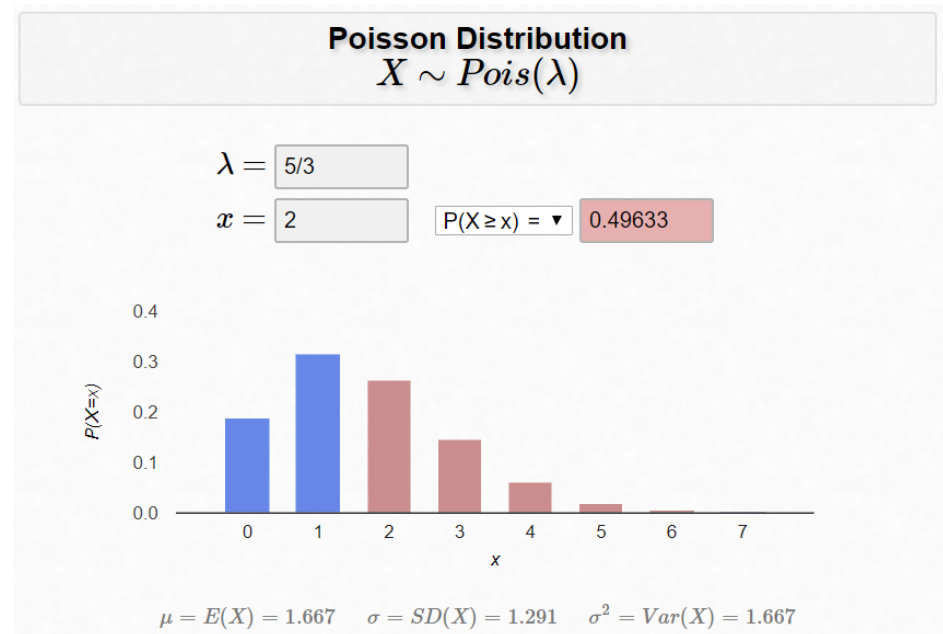
TECHNOLOGY

In calculator

- $1 - \text{poissonpdf}\left(\frac{5}{3}, 0\right) - \text{poissonpdf}\left(\frac{5}{3}, 1\right)$

or

- $1 - \text{poissoncdf}\left(\frac{5}{3}, 1\right)$



EXAMPLE 8

- The monthly worldwide average number of airplane crashes of commercial airlines is 3.5. Find the probability that there will be...
 - At least 2 such accidents in the next month.
 - More than one accident in the next month.

POISSON DISTRIBUTION – EXPECTED VALUE / VARIANCE

- The mean of a Poisson random variable can be computed as follows:

$$\mu = E[X] =$$

- Proof:

POISSON DISTRIBUTION – EXPECTED VALUE / VARIANCE

- Suppose $X \sim \text{POIS}(\lambda)$

- The mean of X is

$$\mu = E[X] = \lambda$$

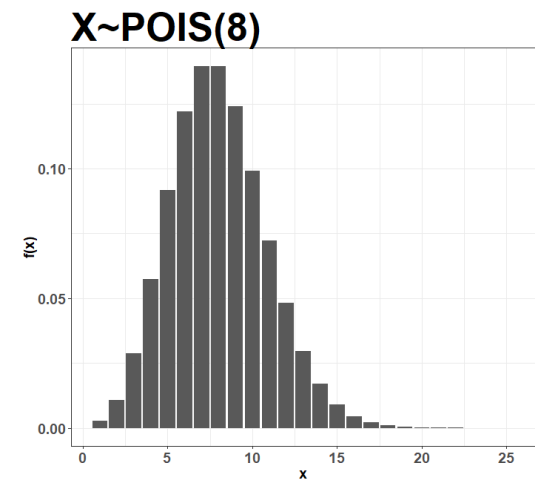
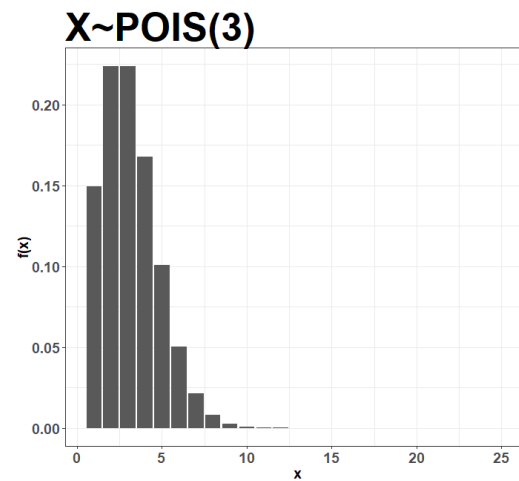
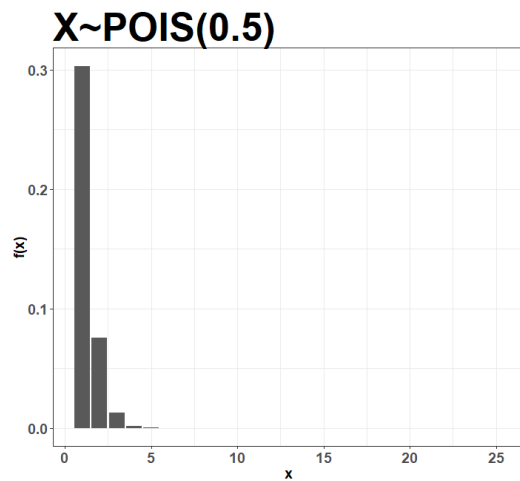
- The variance of X is

$$\sigma^2 = \text{Var}(X) = \lambda$$

- Notice the mean and variance of a Poisson random variable are equal.

POISSON DISTRIBUTION

- What happens as λ increases?



EXAMPLE 9

- Customers arrive at a travel agency at a mean rate of 11 per hour. The agency would like to determine how many agents they need on hand. One agent is needed for every 3 people that walk through the door.
 - Find the mean and variance.
 - Find the probability that at most 3 customers will arrive in a given hour.



OTHER COMMON DISCRETE DISTRIBUTIONS



DISCRETE DISTRIBUTIONS SUMMARY

- **Discrete Uniform Distribution:** equal probabilities
- **Bernoulli Distribution:** one trial
- **Binomial Distribution:** number of successes out of n trials.
- Geometric Distribution: number of trials until the first success
- Negative Binomial: number of trials until the r^{th} success
- Hypergeometric: used to model “finite population” experiments (sampling without replacement)